

LO2.1 and LO2.2: Test Plan and Evaluation

Requirement 1 (R1): Drone paths must never enter no-fly zones.

Priority: This is a safety requirement with high priority, as violations of this requirement lead to unacceptable and unsafe behaviour. The safety critical nature of this requirement justifies allocating testing effort early in the lifecycle.

Pre-requisites:

- Path generation logic must have been implemented
- Restricted area data must be available, or a simulation of the restricted area should be used.

Requirement 2 (R2): Route computation must complete within 30 seconds

Priority: This is a measurable performance requirement with medium priority. While not safety critical, it is an important quality attribute affecting overall system usability.

Pre-requisites:

- The service must be fully integrated, as execution time depends on multiple interacting components.
- Representative workload and input data

Requirement 3 (R3): Drone movement must follow fixed step size of 0.00015 degrees and move in the 16 allowed compass directions

Priority: This is a functional correctness requirement with high priority. It directly affects safety, precision and cost calculations, making early verification important

Pre-requisites:

- TMovement and geometry logic must be implemented
- Validation of angles and position updates must be available

Scaffolding and instrumentation:

R1: Instrumentation will include logging of drone positions while flight path generation is occurring and recording any of these positions that intersect restricted areas. Scaffolding will also be required to use controlled data for restricted areas and synthetic inputs for the path generation logic. This allows paths to be created repeatedly and checked against the safety requirements before full integration.

R2: Timing instrumentation will measure the time it takes to fully compute a path from start to finish. The execution times for the inputs will then need to be logged. For scaffolding, input data must be mocked to provide a realistic example of how the program will be used and to ensure that the performance measurements are repeatable. This allows for performance testing without relying on real data.

R3: Instrumentation will capture successive drone positions and computed angles during movement generation. Step size between consecutive positions will be measured, and angles will be checked

against the allowed compass set. Scaffolding will use deterministic start positions and target points to ensure that movement behaviour is predictable and violations can be detected precisely.

Process and Risk

R1 process: This requirement is addressed at unit and integration test levels, supporting early verification.

Planned activities: Early verification of path generation logic against restricted areas, Integration checks combining path generation with restricted area data.

Lifecycle placement: As soon as path logic and data access are available

Purpose: Detect safety violations early to reduce downstream risk

Risks for R1: If the synthetic restricted area data is unrealistic or incomplete, early tests may provide false confidence. A mitigation for this would be to prioritise accuracy and coverage of synthetic data to reduce this risk.

R2 process: This requirement is addressed at the system test level.

Planned activities: Measurement of end-to-end route computation time with multiple representative and complex inputs.

Lifecycle placement: Late in lifecycle, as system must be fully integrated before tests can be run and produce meaningful results

Purpose: Verify compliance with the specified performance constraint.

Risks for R2: The performance results generated will be entirely dependent on the execution environment and representativeness of synthetic workloads given. The mitigation for this is to use a broad range of inputs and accept the residual performance risk due to project constraints.

R3 process: This requirement is addressed at unit test level mostly, with some integration level checks.

Planned activities: Unit tests verifying valid and invalid movement angles, Boundary tests confirming fixed step size between consecutive positions, Integration checks that exercise the same movement logic indirectly in findPath, while focusing primarily on no fly zone avoidance.

Lifecycle placement: Early, alongside movement and geometry implementation

Purpose: Ensure deterministic, specification compliant movement and prevent subtle errors in safety and cost calculation later on in the lifecycle.

Risks for R3: Rounding may cause small discrepancies in numbers, but this can be mitigated by using tolerance based assertions aligned with the specification's precision rules

Evaluation: This test plan focuses on a small number of high value requirements and allocates testing effort according to risk, measurability and feasibility. It ensures that the safety critical behaviour and movement rules are verified early on, while leaving performance testing until a full meaningful measurement can be made. Although limited in scope, the plan demonstrates prioritisation, lifecycle-aware scheduling, and explicit identification of constraints and residual risks.